FALL 2026 | MEE4033

Mechatronics

Mechatronics (MEE4033) · Department of Mechanical Engineering

Lecture 13

Motor Control 2: DC Motor Modeling

Mechatronics with Experiments, Sabri Cetinkunt, Wiley (Ch. 8)



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Basic Electric Equations

Ideal resistor

$$V(t) = R\,I(t)$$

- **V(t)**: voltage across the resistor
- **I(t)**: current through the resistor
- **R**: resistance

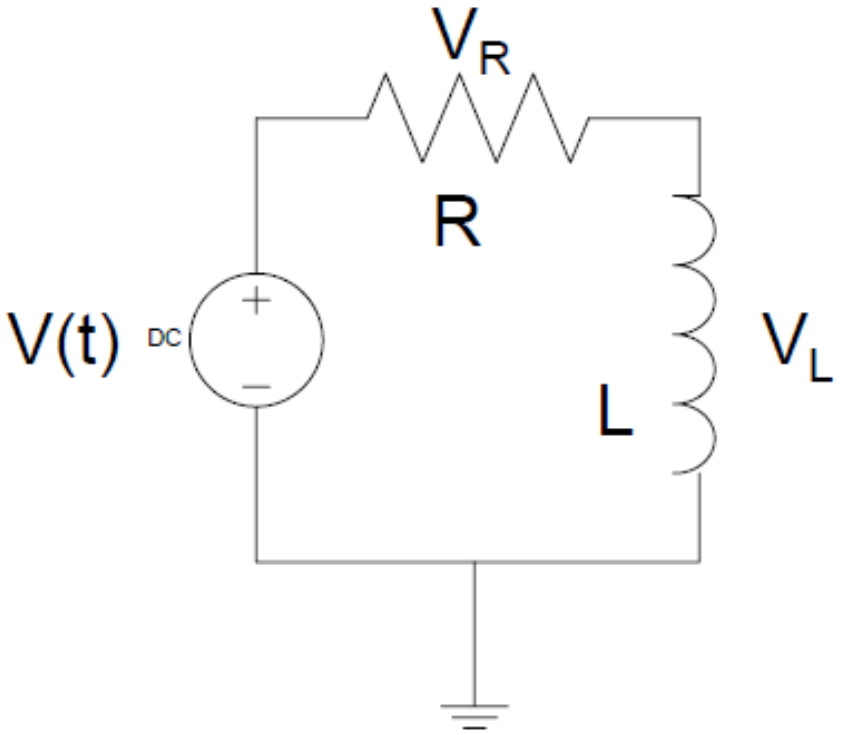
Ideal inductor

$$V(t) = L\,\frac{d}{dt}I(t)$$

- **L**: inductance
- The voltage is proportional to the **rate of change** of the current, not to the current itself

Basic Electric Equations: Kirchhoff's Voltage Law

By Kirchhoff's voltage law



$$V_L(t) + V_R(t) - V(t) = 0$$

$$V(t) = R\,I(t) + L\,\frac{d}{dt}I(t)$$

DC Motor Principle: Lorentz Force and Faraday's Law

Lorentz force

- In a steady magnetic field
  - Constant magnetic flux density, ***B***
  - Current through the conductor, ***I***
  - Length of the conductor, ***L***

$$\vec{F} = I\vec{L} \times \vec{B}$$

Faraday's law of induction

- Electromotive force (EMF) voltage, E_g
- Velocity of the conductor, ***v***

$$E_g = (\vec{v} \times \vec{B}) \cdot \vec{L}$$

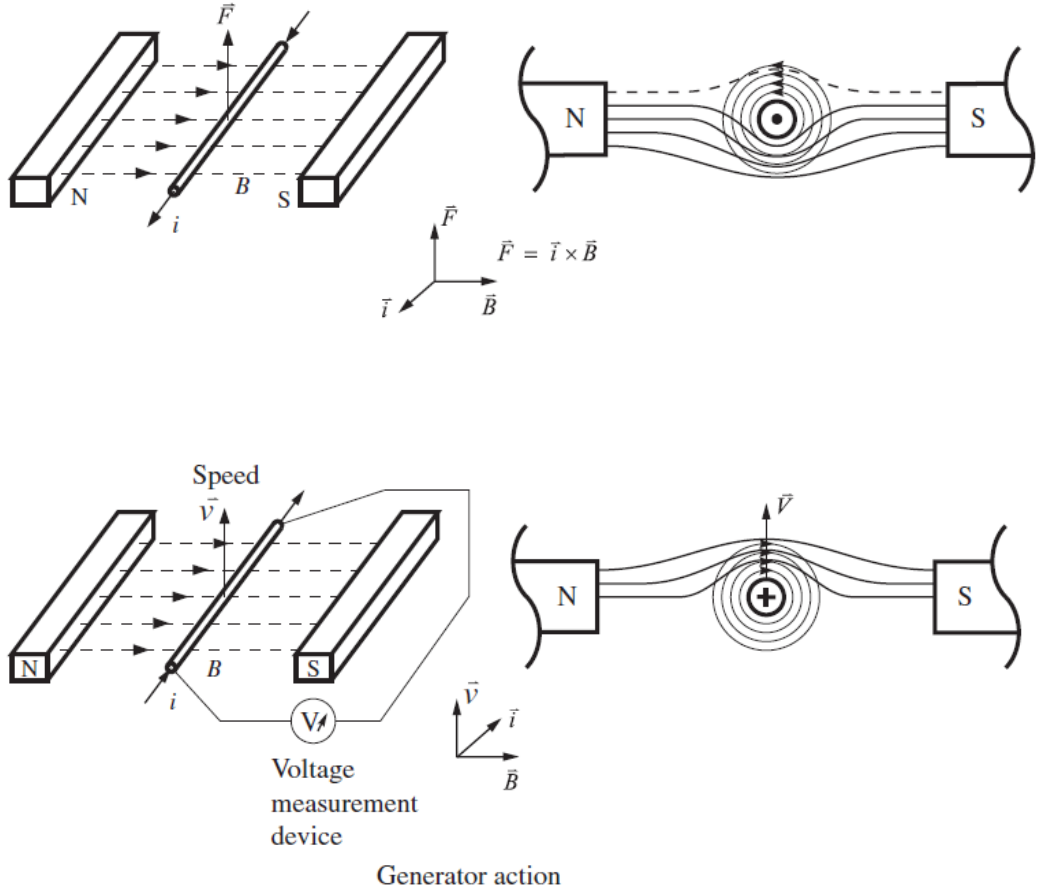


FIGURE 8.26: DC motor operating principles: a current carrying conductor in a magnetic field.

Torque Generation in a Motor

- If the current-carrying rod is **off-centered**, a torque is generated in the rotor

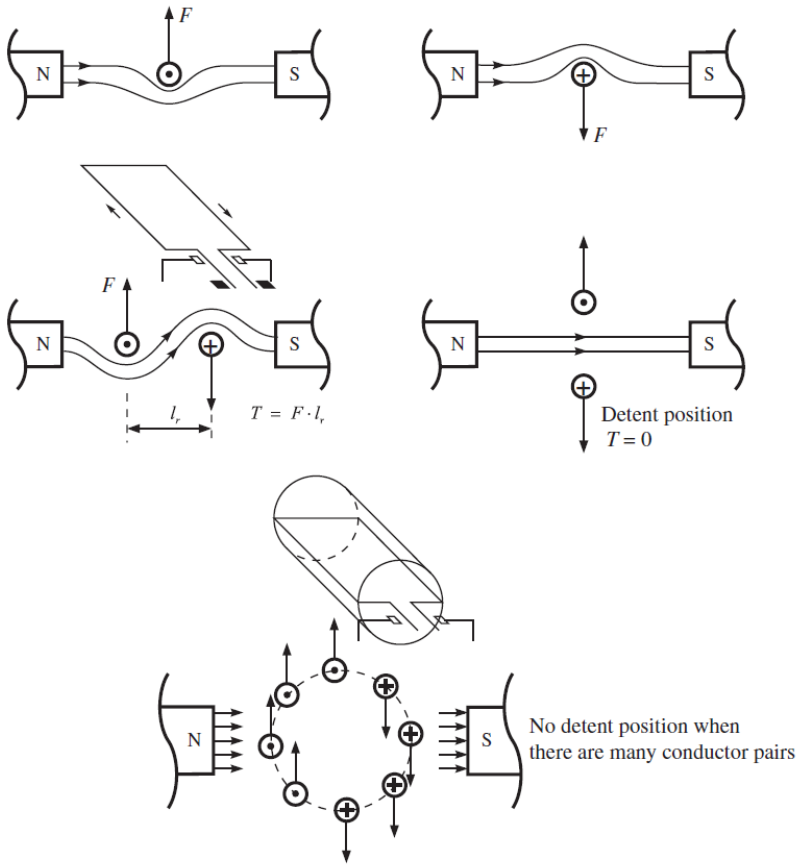
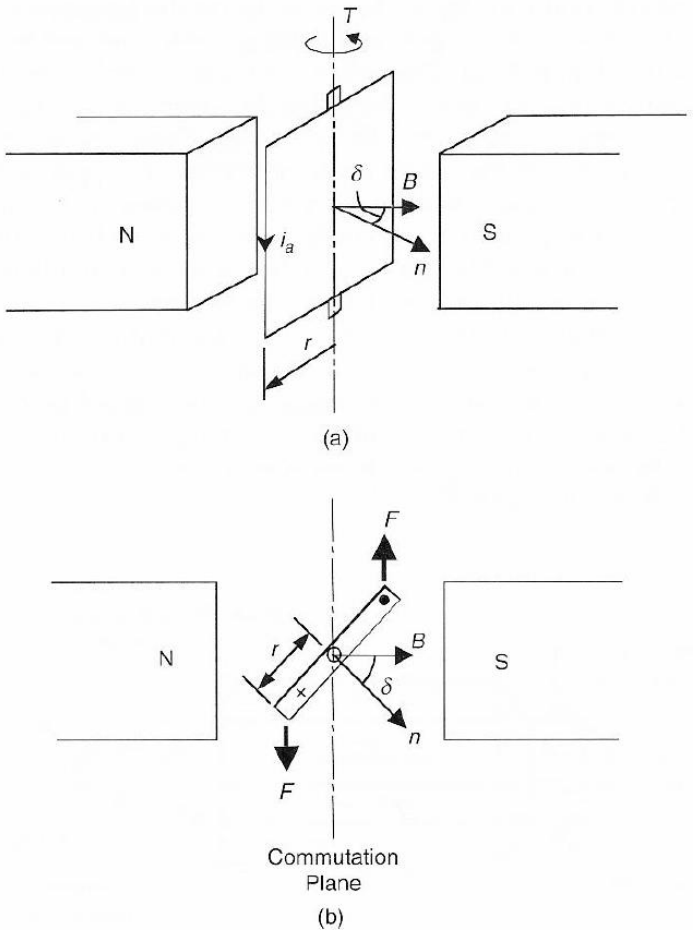


FIGURE 8.27: DC motor operating principles: obtaining a continuous torque by winding a coil around a rotor.

Torque Generation in a Motor: Torque of One Coil

- If the current-carrying rod is off-centered, the torque generated in the rotor is given by

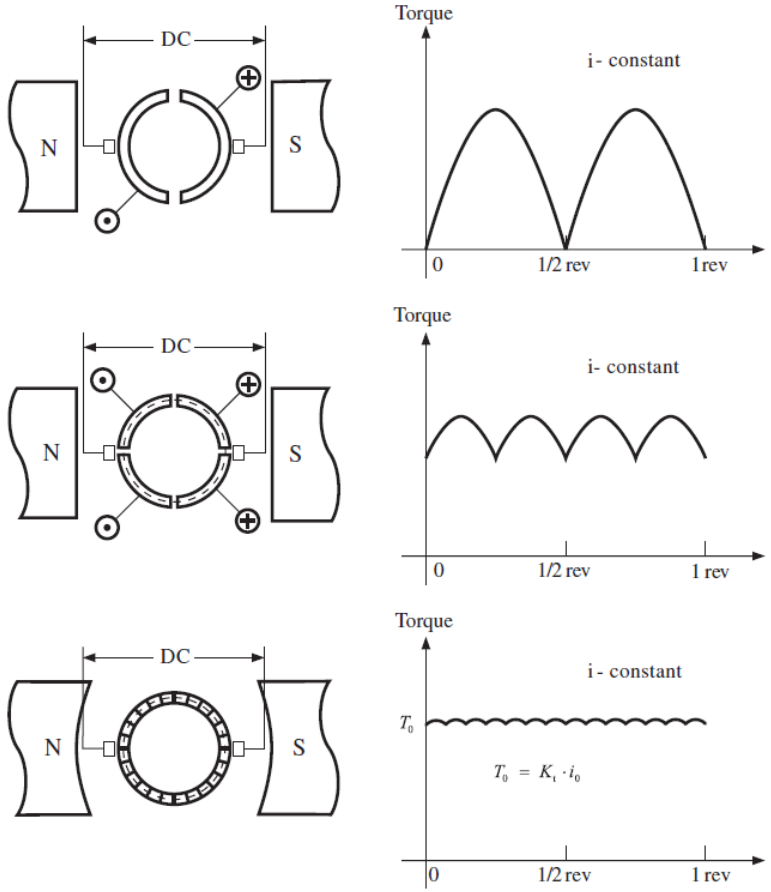
$$T_g = F \times 2 \times r \sin\delta$$
$$= BIL \times 2 \times r \sin\delta$$



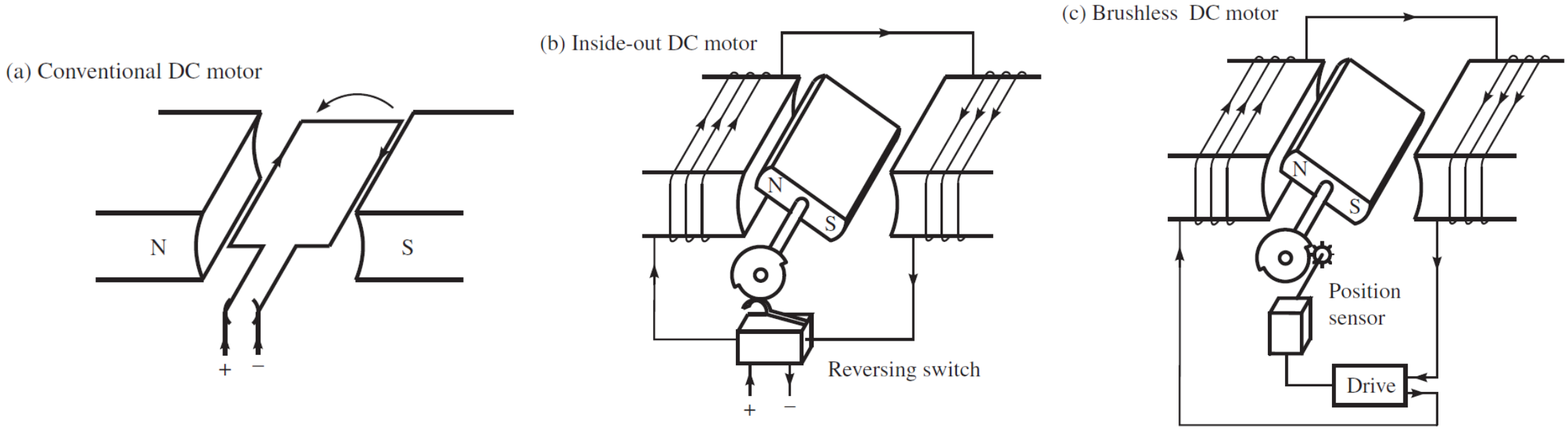
- ***r***: radius of the rotor (lever arm), ***δ***: angle between the coil plane and the field
- The factor 2 counts the two sides of the coil; the torque varies with sin δ as the rotor turns

Torque Generation in a Motor: Torque Ripple

- **Torque ripple** magnitude and frequency are a function of the number of commutations (coils) per revolution

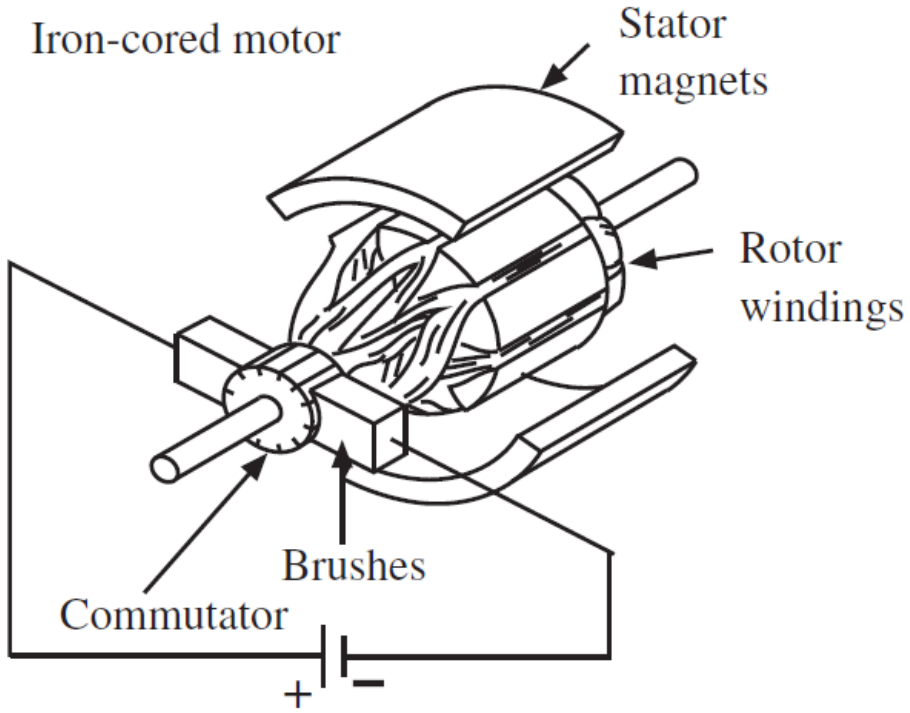


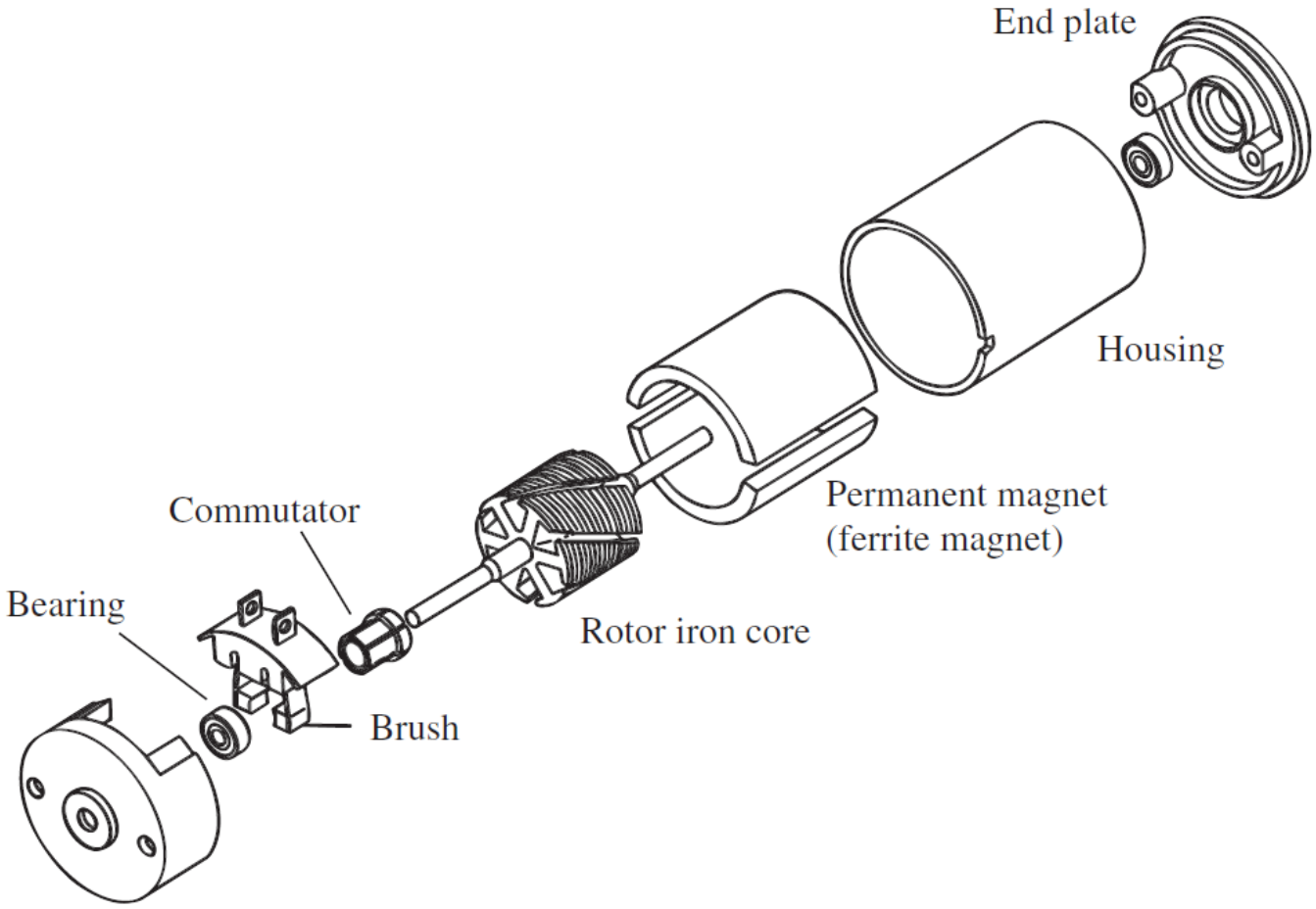
Torque Generation in a Motor: Motor Configurations



- **(a) Conventional DC motor:** stationary magnets, rotating coil, mechanical commutation (brushes)
- **(b) Inside-out DC motor:** rotating magnets, stationary coil; current reversed by a switch
- **(c) Brushless DC motor:** a position sensor and a drive circuit replace the mechanical commutator

Structure of a Brushed DC Motor





Electric Circuit of a Brushed DC Motor Drive

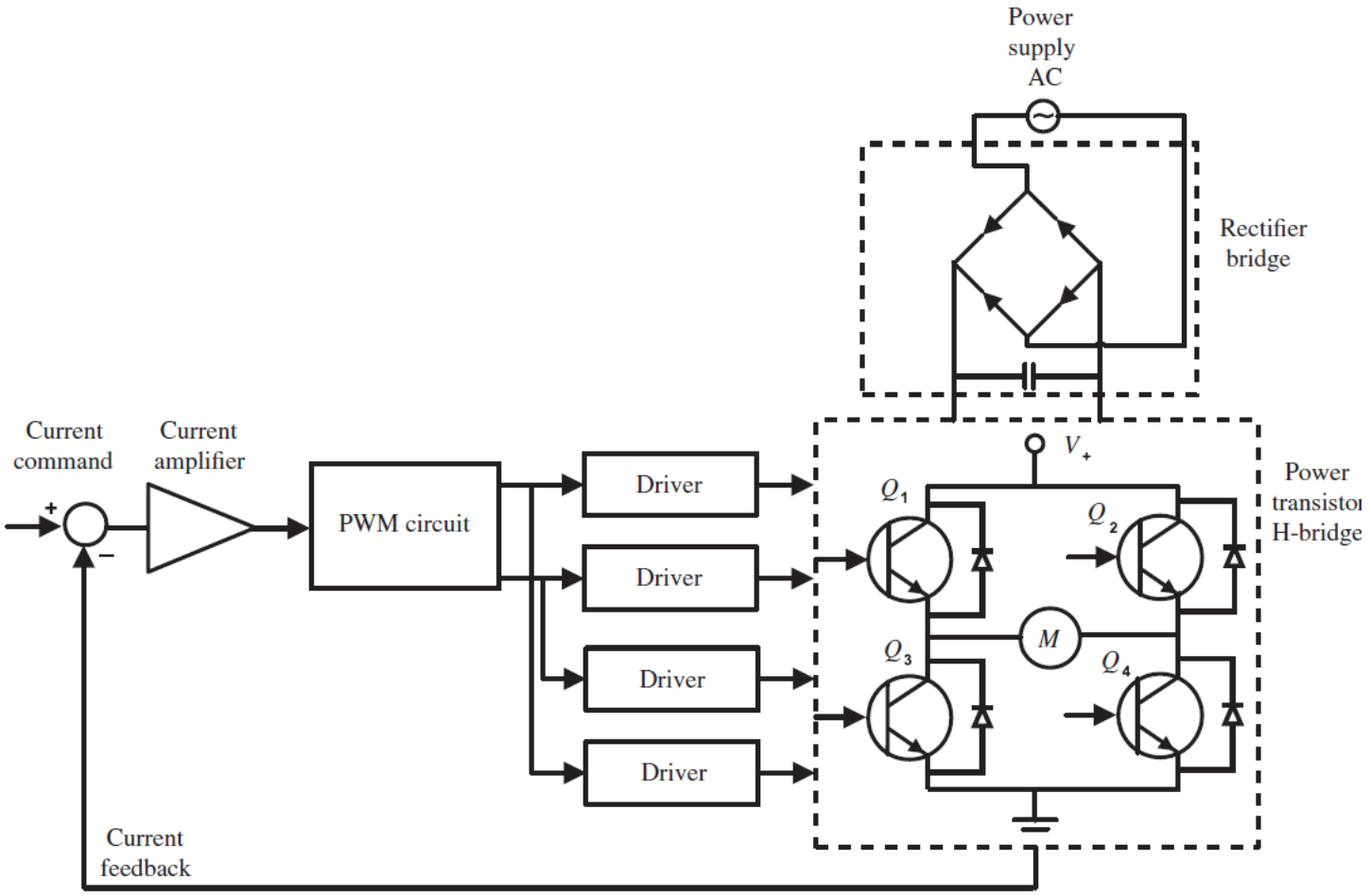


FIGURE 8.30: Block diagram of the brush-type DC motor drive: PWM amplifier with current feedback control.

DC Motor Equation: Motor Constants

- For a permanent-magnet DC motor, the torque T_g generated by the motor is linearly related to the armature current I_a by

(K_T: torque constant)

$$T_g = K_T I_a$$

- The induced back e.m.f. E_g is

($1/K_E$: speed constant)

$$E_g = K_E \omega$$

- **K_E** and **K_T** are motor constants that depend on
  - Rotor dimension (geometry)
  - Armature construction (number of turns in the armature winding)
  - Magnetic field (magnet strength)

DC Motor Equation: K_T and K_E Are the Same Constant

- In the case of **ideal** energy conversion:
  - (general law: the applied electrical power is converted to mechanical power without loss)

$$T_g \, \omega = E_g \, I_a$$

- And therefore the motor constants satisfy

$$K_E \ (\mathrm{V/rad \cdot s^{-1}}) = K_T \ (\mathrm{N \cdot m/A})$$

(Be careful about the unit!)

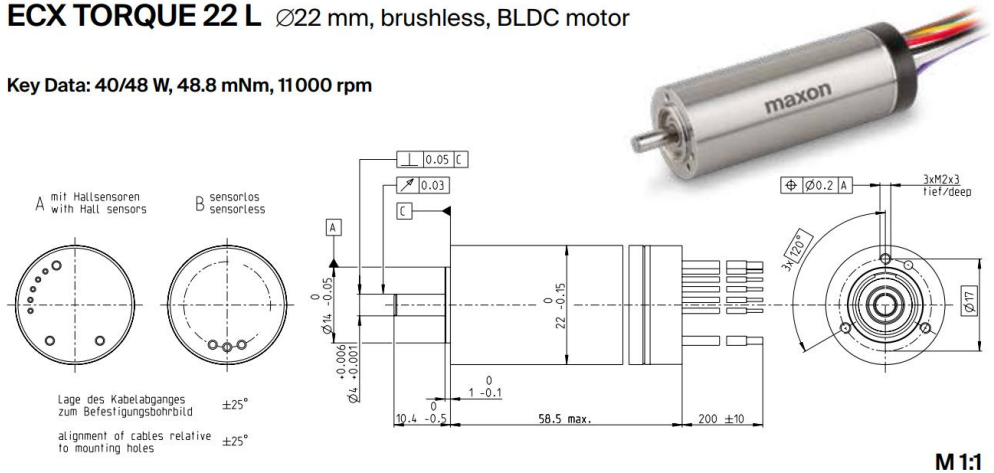
DC Motor Equation: Reading the Constants from a Datasheet

[maxon ECX TORQUE 22 L datasheet (maxongroup.com, EN-21-226.pdf)]

ECX TORQUE 22 L ∅22 mm, brushless, BLDC motor

Key Data: 40/48 W, 48.8 mNm, 11000 rpm

ECX TORQUE



	Motor Data						
1_	Nominal voltage	V	12	18	24	36	48
2_	No load speed	rpm	9380	10300	10300	10700	10900
3_	No load current	mA	224	172	129	90.7	69.9
4_	Nominal speed	rpm	7930	8840	8850	9190	9380
5_	Nominal torque (max. continuous torque)	mNm	45.7	46.2	48.1	45.2	48.8
6_	Nominal current (max. continuous current)	A	3.7	2.74	2.13	1.39	1.14
7_	Stall torque	mNm	528	629	684	644	766
8_	Stall current	A	43.6	38.1	31.1	20.2	18.3
9_	Max. efficiency	%	86.4	87.2	87.7	87.2	88.2
10_	Terminal resistance	Ω	0.275	0.473	0.771	1.78	2.62
11_	Terminal inductance	mH	0.148	0.275	0.49	1.03	1.77
12_	Torque constant	mNm/A	12.1	16.5	22	31.9	41.8
13_	Speed constant	rpm/V	789	579	434	299	228
14_	Speed/torque gradient	rpm/mNm	17.9	16.6	15.2	16.7	14.3
15_	Mechanical time constant	ms	0.575	0.531	0.488	0.536	0.459
16_	Rotor inertia	gcm²	3.06	3.06	3.06	3.06	3.06
	Thermal data			**Operating Range**			

DC Motor Equation: Electrical Model

Armature circuit

$$V_a = L_a \frac{dI_a}{dt} + R_a I_a + E_g$$

- V_a: supply voltage to the armature
- L_a: inductance of the armature winding
- R_a: resistance of the armature winding
- E_g: back e.m.f. generated by the rotation

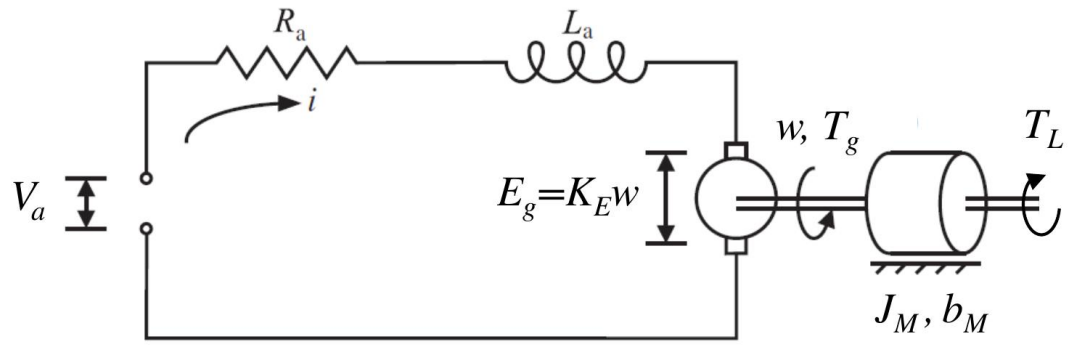


FIGURE 8.59: DC motor dynamic model.

DC Motor Equation: Electrical Model (Neglecting L_a)

- In most cases the inductance can be ignored because the electrical time constant (L_a/R_a) is usually ***much lower*** than the mechanical time constant given by the rotor inertia
- In **steady state**, the effect of L_a is zero

$$V_a = L_a \frac{dI_a}{dt} + R_a I_a + E_g$$

$$\Downarrow \quad L_a \rightarrow 0, \quad E_g = K_E \omega$$

$$V_a = R_a I_a + K_E \omega$$

Torque-Speed Characteristic Curve

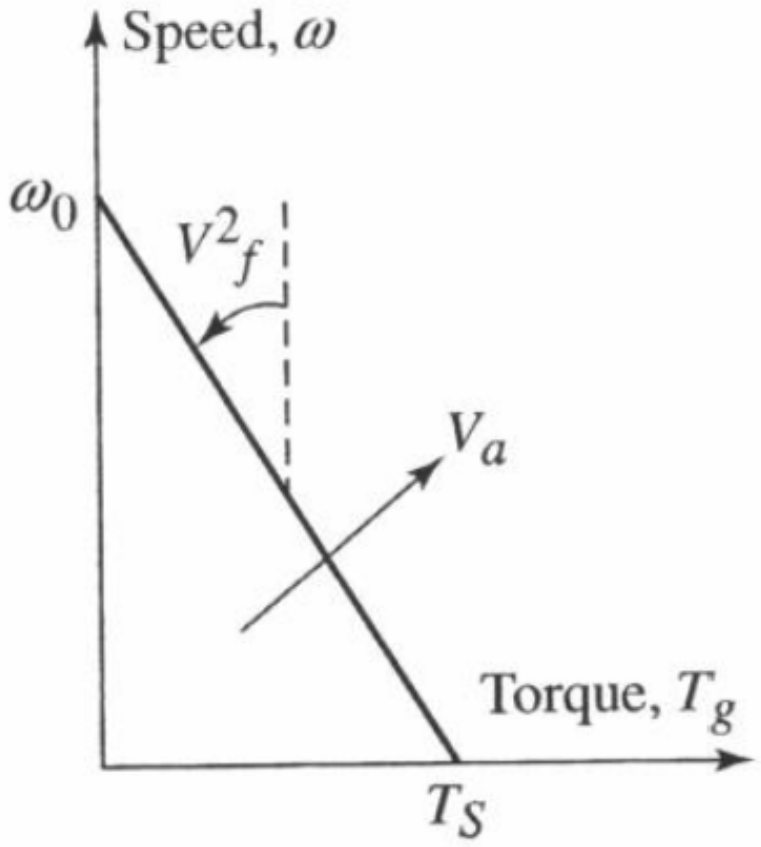
- Performance of a DC motor
  - Stall torque
  - No-load speed
  - Constant speed behavior

$$V_a = R_a I_a + K_E \omega$$

$$= R_a \frac{T_g}{K_T} + K_E \omega$$

$$\Rightarrow \quad T_g = \frac{K_T}{R_a} V_a - \frac{K_T K_E}{R_a} \omega$$

$$= \frac{K_T}{R_a} V_a - K_M^2 \omega$$



Steady state torque-speed characteristics

Torque-Speed Characteristic Curve: Stall Torque and No-Load Speed

Stall torque

- By setting $\omega = 0$,

$$T_s = \frac{K_T}{R_a} V_a$$

- The blocked-rotor torque at the applied voltage V_a

No-load speed

- By setting $T_g = 0$,

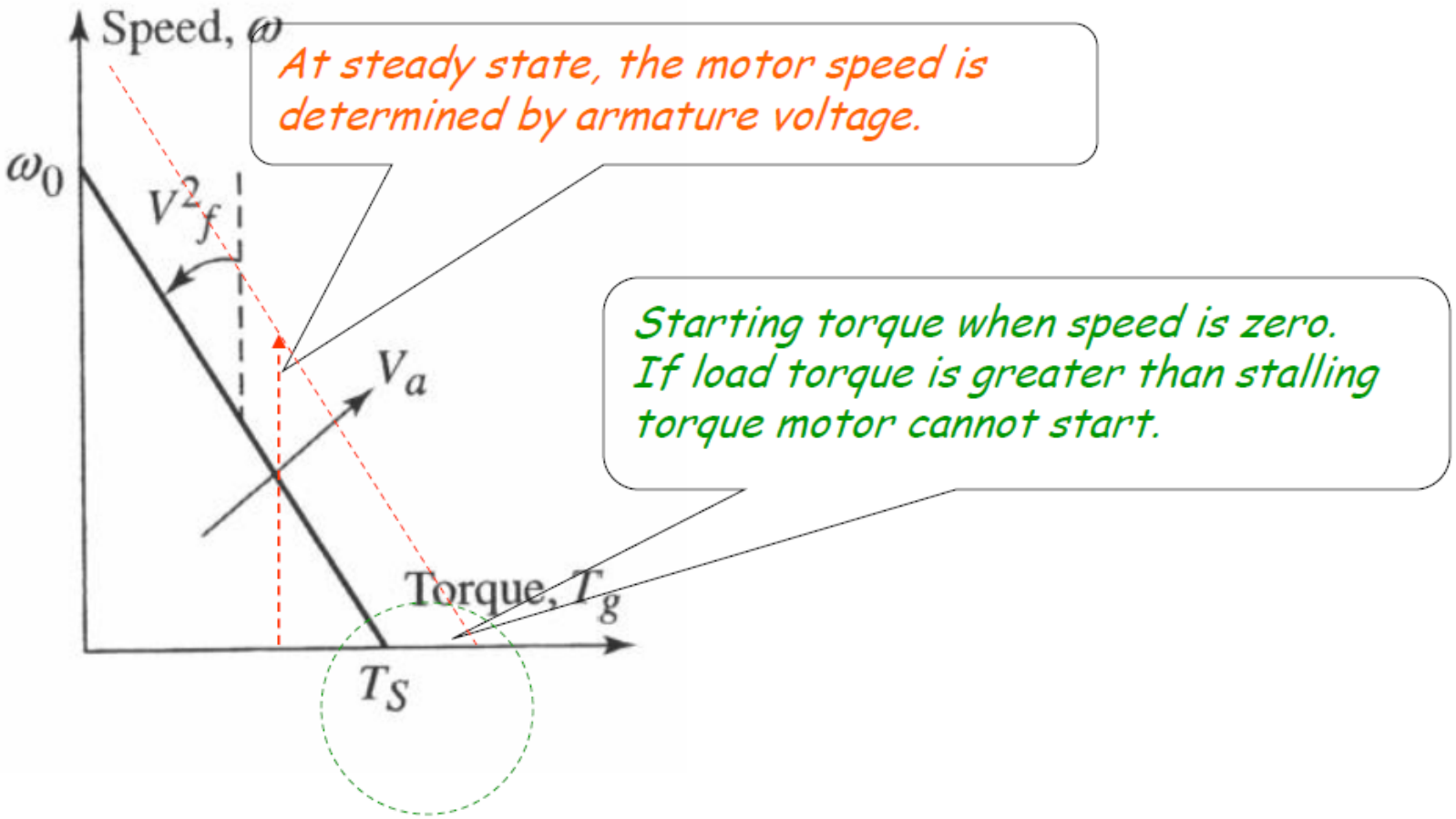
$$\omega_0 = \frac{V_a}{K_E}$$

- The speed at which the back e.m.f. equals the applied voltage

- Both endpoints of the line are set by V_a: a stronger motor (larger K_T, smaller R_a) has a larger stall torque, and the no-load speed depends only on K_E

Torque-Speed Characteristic Curve: Effect of V_a

- If V_a is increased, the curve translates to the right (parallel shift, same slope)



DC Motor Equation: Mechanical Model

Applying Newton's law to the rotor

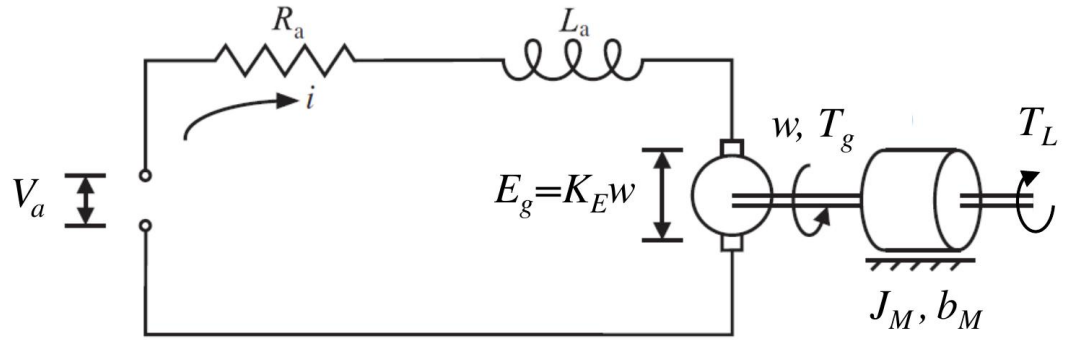


FIGURE 8.59: DC motor dynamic model.

$$T_g = J_M \frac{d\omega}{dt} + T_L + b_M \omega$$

- J_M: moment of inertia of the rotor
- b_M: viscous damping constant of the motor
- T_L: load torque

DC Motor Equation: Putting All Together

- We can ignore the small L_a and assume there is no load attached ($T_L = 0$)

$$V_a = R_a I_a + K_E \omega$$

$$I_a = \frac{V_a - K_E \omega}{R_a}$$

$$\begin{aligned} J_M \dot{\omega} + b_M \omega &= T_g - \cancel{T_L} \\ &= T_g \\ &= K_T I_a \\ &= K_T \frac{V_a - K_E \omega}{R_a} \end{aligned}$$

DC Motor Equation: Final Form

Voltage - angular velocity equation

$$J_M\dot{\omega} + \left(b_M + \frac{K_T K_E}{R_a}\right)\omega = \frac{K_T}{R_a}V_a$$

friction damping

back e.m.f. damping
generated by motor itself

- A **first-order** linear system from V_a to ω: the back e.m.f. adds damping $K_T K_E / R_a$ on top of the friction b_M